JACK PHILPOTT

GIS & Data Specialist | Humanitarian information management & geospatial analysis

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PROFILE

GIS and data specialist with 7 years' experience in humanitarian and environmental information management across REACH, ICRC and FAO, working on Nigeria, Ukraine, South Sudan and global programmes. Technical lead and focal point for multi-sector teams, supporting many projects at once across health, WASH, food security, nutrition, protection, cash and markets, and climate. Builds end-to-end data systems in R, Python, SQL and ArcGIS/QGIS: survey tools, automated and AI-assisted cleaning pipelines, geospatial and remote-sensing analysis, cloud data warehouses and dashboards. Has led GIS and data teams of up to six.

EXPERIENCE

Senior Research Officer | IMPACT Initiatives (REACH) — Jun 2026 – Present

Nigeria mission, based in Dakar, Senegal

- Led the Nigeria Multi-Sector Needs Assessment (MSNA), covering 32,000 households across 11 states and 176 LGAs of northern Nigeria, from methodology and tool design through data collection, analysis and reporting.
- Designed a tailored sampling methodology that kept coverage across a large area of northern Nigeria despite severe, widespread access constraints from insecurity.
- Led and supported a project team of three, setting priorities and coordinating work across the research cycle.
- Built an automated system in R and Python to process and quality-check up to 1,000 survey submissions a day, with a live monitoring dashboard and guidance documents for 20+ partner organisations.
- Integrated AI tools into code development and pipeline automation, and into drafting presentations and partner guidance, in line with data protection requirements.
- Presented results to Nigeria's Cadre Harmonisé (IPC-compatible food security analysis), informing humanitarian programming and decision-making.

GIS & Data Specialist | International Committee of the Red Cross (ICRC) — Feb 2025 – Feb 2026

Analysis & Evidence unit, Kyiv, Ukraine

- Served as the delegation's GIS and data focal point and represented ICRC in humanitarian working groups, supporting economic security, health, protection, water and habitat, forensics and security programmes.
- Managed a team of three GIS and data staff, providing day-to-day supervision, technical mentoring and capacity building.
- Oversaw data collection across the delegation (KoboToolbox, DeviceMagic), covering data quality, accuracy and security, and trained programme and operations staff.
- Developed remote-sensing workflows (ArcGIS Pro, Google Earth Engine) for flood monitoring, crop detection and monitoring of critical sites near the frontline, including power infrastructure, nuclear power plants and cemeteries, supporting the safety of staff and communities in areas with restricted access.
- Built geo-analytical models in R and Python, and dashboards, factsheets and story maps in Power BI and ArcGIS (Dashboards, Experience Builder, StoryMaps).
- Led project management of the unit's information products and data systems using SCRUM on Azure DevOps.

Career break: Spanish language study and personal programming and GIS projects — Apr 2024 – Jan 2025

Geospatial Analyst | Food and Agriculture Organization of the UN (FAO) — Apr 2023 – Mar 2024

Livestock Information, Sector Analysis and Policy Branch, Rome, Italy

- Ran country, regional and global analyses of livestock emissions in R using the Global Livestock Environmental Assessment Model (GLEAM).
- Supported countries with mitigation scenarios and situation reports, presented findings to government stakeholders, and delivered three Training of Trainers workshops on data collection, analysis and mitigation strategies.
- Designed a standardised geospatial data warehouse on Google Cloud (BigQuery, Earth Engine, Cloud Storage), using SQL and JavaScript to integrate multi-format data and feed Looker Studio and R Shiny dashboards.
- Built dashboards and factsheets in Power BI, R Shiny and QGIS, and ran R analysis on remote Linux servers over SSH.
- Contributed analysis, visualisation and writing to annual reports, and represented FAO at international conferences.

Career break: travel and Spanish language study — Oct 2022 – Mar 2023

GIS Specialist | IMPACT Initiatives (REACH) May 2019 – Sep 2022

Juba, South Sudan · Promoted three times: Junior GIS Officer → GIS Officer → Senior GIS Officer → GIS Specialist

- Led all GIS and data workflows for the South Sudan mission, covering the MSNA, Integrated Needs Tracking, Area of Knowledge, Joint Market Monitoring Initiative (JMMI), port and road monitoring, population movement baseline, WASH infrastructure and climate monitoring, and supporting IPC analyses.
- Managed a multicultural team of six GIS and data staff (two national, four international), and set up a fellowship programme mentoring two national GIS fellows.
- Supported every stage of the research cycle: KoboToolbox survey design, data management and cleaning, analysis, dashboards, web GIS applications, maps and factsheets.
- Represented REACH in the Information Management Working Group, helping OCHA chair meetings and supporting cluster IMOs with data needs, including People in Need (PiN) calculations for the HNO.
- Provided regional support to five East African country offices, supported sudden-onset emergency responses, and contributed to global GIS and data development.
- Developed and ran a climate monitoring system for over two years, using Python in Google Earth Engine with CHIRPS rainfall, NDVI and satellite flood-extent data to track flood and drought risk across South Sudan and East Africa. Its monthly data, quarterly factsheets and dashboard were used by OCHA, clusters and partners; led three post-flood field assessments, including ground verification.
- Produced early-warning and anticipatory action reports, analysing long-term historical and geographic trends to identify areas likely to flood again and estimate the potential impact.

GIS Intern | IMPACT Initiatives (REACH) Nov 2018 – May 2019

Kyiv, Ukraine

- Supported MSNAs, rapid and household assessments, and network analysis of internally displaced persons' movements.
- Cleaned and aggregated spatial and non-spatial data, and produced thematic maps (ArcMap, Illustrator) and web apps (ArcGIS Web AppBuilder, R Shiny with Leaflet).
- Trained partners to use the spatial components of their assessment data.

GIS Technician | Cyient Jul 2016 – Feb 2017

For DEFRA and the Rural Payments Agency, UK

- Digitised and mapped farmland from aerial and satellite imagery, spectral crop-type analysis and ground-truth surveys, informing EU Basic Payment Scheme subsidies to UK farmers.

SKILLS

Programming: R (primary), Python, SQL, JavaScript, HTML/CSS, Linux/SSH

AI & automation: AI-assisted code development (R, Python), workflow automation and drafting of reports, presentations and guidance; responsible use with sensitive data

GIS & remote sensing: ArcGIS Pro (ArcPy), QGIS, Google Earth Engine, FME, Mapbox, PostgreSQL (pgAdmin)

Data engineering: Google Cloud (BigQuery, Cloud Storage), automated cleaning and QA pipelines, Azure DevOps (SCRUM)

Visualisation: Power BI, R Shiny, Tableau, ArcGIS Dashboards, Experience Builder and StoryMaps, Adobe Illustrator and InDesign

Methods: Spatial and statistical modelling, remote sensing, multi-criteria decision analysis (AHP), machine learning, cartographic design

Data collection & research: KoboToolbox, DeviceMagic, survey design, sampling, MSNA, IPC and Cadre Harmonisé

EDUCATION

MSc Geographic Information Science (Merit) | University College London (UCL) 2017 – 2018

- Thesis with WSP: multi-criteria (AHP) suitability model of rail overbuild opportunities in London, delivered as an interactive ArcGIS/Python decision-support tool.

BSc Physical Geography | Swansea University 2013 – 2016

- Dissertation: glacier terminus retreat and climate response in the Hayes Range, Alaska, measured from satellite imagery.

LANGUAGES & TRAINING

Languages: English (native) · Spanish (basic conversational) · French (beginner, currently studying)

Training: HEAT (hostile environment security training) · Data protection · PSEA